

No. of Functional Features Included in the Solution

Date	05 July 2026
Team ID	XXXXXX
Project Name	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Voice Input Upload	Allows users to upload voice explanation audio files	Input Handling Module	Done	High
2	Speech-to-Text Conversion	Converts user speech into text format for analysis	Whisper STT Module	Done	High
3	Concept Understanding Analysis	Evaluates user's explanation using semantic similarity	NLP Analysis Engine	Done	High
4	Audio Feature Evaluation	Analyzes speech features and quality of audio	Audio Processing Module	Done	Medium
5	Score Generation	Generates performance score based on analysis results	Scoring Engine	Done	High
6	Dashboard Visualization	Displays results using charts and progress indicators	Streamlit UI Module	Done	Medium
7	PDF Report Generation	Creates downloadable feedback report for users	Report Module	Done	High
8	Cloud Deployment	Hosts application online for user access	Streamlit Cloud	Done	Medium

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	5

Additional / Nice-to-Have Features	3
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Voice upload interface, Dashboard visualization
2	Backend / Logic	3	Speech conversion, Concept analysis, Score generation
3	Database / Storage	1	Audio files and generated report storage
4	API / Integration	2	Whisper model, Sentence Transformer integration
5	Security / Authentication	1	Session handling and secure file processing